

Simulation of alternative regulations in the Colombian electricity market[☆]

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Abstract

The deregulation of the Colombian electricity system took place in 1994 and the pool started operations in 1995. The Colombian system adopted a capacity charge mechanism to increase incentives to invest in new capacity. The capacity charge was showing strength, and apparently driving investments during the initial years. However, the mechanism started to exhibit weaknesses in terms of transparency and disincentives, causing a negative effect on investments. Different authors have presented alternative regulatory options to update the system. A non-standard system dynamics approach to evaluate alternative regulation schemes for the Colombian electricity market is proposed. A specific regulation problem is undertaken to illustrate the proposed methodology. It shows how the capacity charge mechanism, which has been used for reliability purposes, might be changed for alternative schemes. The proposed transformations to the actual regime seem to overcome some of its drawbacks. Simulation results indicate these alternatives improve the general system behaviour. In addition, the underlying model has been used afterwards for other energy policy purposes.

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1. Regulating the new electricity industries

The electricity industry has been transformed worldwide since the mid 1980s. The deregulation process has replaced state-owned monopolies with open and competitive markets in those parts of the supply chain where it seemed feasible, taken elements from other deregulated industries such telecommunication and airlines. These have prompted a new paradigm for the activities of trading, management and delivery of energy services to final customers as well as for regulation [2]. Chile was the first country to move ahead in this field in the 1980s, followed by the UK and Norway in the early 1990s, with major innovations in market openness, privatisation schemes and regulation set-ups. In Latin American, most countries have undertaken major reforms or are in the process of transforming their electricity industries; in particular Colombia, whose pool

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started operations in 1995 [3]. Both the evolution and the worldwide spread of the new industry paradigm have been very rapid.

The underlying motivations of deregulation have been to provide electricity more efficiently and reliably with higher quality; as well as regional needs, in order to prevent mismanagement in some Latin American countries and coal dependence in the UK [4,5]. The industry is complex and uncertain, with no room for generalisation about the optimum solution for the electricity industry scheme. On one hand, some benefits are already being perceived, such as progress in technology and reduced prices [6]. On the other hand, the systems are still far from reaching equilibrium and major reforms are underway in many countries. Limitations in adaptation have created problems such as the well-known Californian case [7] and the Chilean Crisis [8], where major outages and extremely high electricity prices have been observed due to regulatory problems.

We have focused on the Colombian electricity market (CEM). The deregulation took place in 1994 with laws 142 and 143 [9,10], and the pool started operations in July 1995. During the 1st year, the system developed satisfactorily in some sectors, especially in generation, sales to large customers, and recently competition has also reached the household sector [6]. The Colombian system adopted a British-like structure by adding a capacity charge (CC) mechanism to entice investments in new capacity. The CC was showing strength and apparently driving investments during the initial years. However, the mechanism started to exhibit weaknesses in terms of transparency and disincentives, with negative effects on investments.

Given the current situation, there is a call for a new round of reforms to update the market rules, in order to avoid undesirable blackouts, as is also suggested by other reports [6,11–14]. Two types of alternatives—regarding the regulator involvement—have been proposed. The first is an administrative solution, where the regulator has an important role in guiding the market. The second is a market-oriented solution, where the regulator takes a less interventionist role. In this paper we simulate some of the proposed alternative regulatory options in the CEM. We present the simulation model which includes, among other factors, the behaviour of the investors under the proposed market schemes. We discuss two of seven alternative regulatory options: (i) an administrative solutions presented in [14], and (ii) the creation of an option market proposed with variations in different reports [11,13,14].

The following section briefly presents the modelling approach chosen for analysis, a short description of the CEM and the system dynamics platform for assessing those regulation alternatives. Next, we explain and assess the two selected alternative options that may be implemented in the CEM. Finally, we present our conclusions in the light of the simulation process and results. We observe that the implementation of the alternative with an option market helps to improve the reliability of the electricity supply.

2. The modelling platform for the assessment of alternative regulatory options in CEM

From a regulatory perspective, the electricity industry has been dominated by a number of uncertainties at all levels, so stability has been an important objective. The regulator's task is far from simple. The regulator needs to provide what the market is lacking and should strike a balance between all stakeholders and have special concerns for two of them: suppliers and customers. From a simplified view, the regulator may pursue goals to attain intensity in competition so that prices will reflect the incremental industry cost, with clear signals for effective system expansion to satisfy demand at competitive prices and good electricity quality. In such environment, learning from reality may be inappropriate, causing catastrophic effects for the whole system, because of the long lags involved in this industry. Learning from models might help to assess the likely consequences, so the policy maker may investigate alternatives to eliminate undesirable effects [15].

Electricity markets face uncertainties and complexities which create difficulties for modelling; where optimisation of the system seems to be an unfeasible alternative, behavioural methodologies are more likely to address the problems. Modelling causality and delays become an important practise to account for policy effects on the system. In particular, this modelling may indicate the need for continuous adjustment of the regulatory framework in order to obtain the desired system behaviour. Modelling causality and delays also help investigate whether policies trigger instabilities which may affect system performance. System dynamics modelling is a tool which incorporates these elements.

The system dynamics school of modelling has been used to explore different policy issues during the last decades [16–27]. With a system dynamics model, it will be possible to analyse and evaluate the behaviour of

the system under alternative regulatory options. In fact, electricity markets have caught the attention of a number of scholars and consultants, who use a variety of tools for analysis. These tools include simulation and risk analysis techniques, for investment purposes, optimisation approaches at the operational level, and both simulation and econometric techniques for the evaluation of regulation alternatives [28]. System dynamics modelling mainly focuses on trying to understand how structure influences behaviour in terms of (i) its dynamics (because of difficulties in forecasting non-linearities and delays), (ii) instabilities (due to the dynamics resulting from the interrelationships between variables and initial conditions), and (iii) counter-intuitive behaviour (as a result of the interrelationships between variables and delays).

An electricity system depends on its generation, transport and supply characteristics. It possesses underlying dominant structures. Behaviour is influenced by physical, technological and topological considerations. Other softer structural aspects such as institutions, policies and regulations have both short-term effects on behaviour and a long-term impact on the physical structure of the system, which reinforces or transforms system evolution. Fig. 1 indicates the main macro structure dynamics of such electricity structures. In this case, the system dynamics model takes the form of a system dynamic platform for policy analysis as explained in [20].

Market behaviour is determined by both physical and market structures. The physical structure represents the topology of the system, which includes the installed capacity, the main constraints, gas availability, etc. The market structure represents the institutions and market rules. Agents take actions such as investment decisions and bidding in the pool under a given market structure and topology of the system. Investment decisions change the physical structure of the system after a delay. During that time, the regulator observes the behaviour of the system and the physical condition in order to change the market structure. The change in market structure represents the application of alternative regulatory options.

The evaluation of each alternative regulatory option over time consists of establishing the evolution of the physical and topological structure. This evolution depends on investment decisions with respect to capacity, based on a real options approach. In this way, each alternative is assessed with respect to its reliability and price to customers, including volatility. Modelling to evaluate the regulatory schemes was also proposed by TERA [13] to the Comisión de Regulación de Energía y Gas—Electricity and Gas Regulation Body in Colombia (CREG).

Next, a brief description of the CEM is illustrated, followed by a description of the system dynamic platform for policy analysis.

2.1. The CEM

The characteristics of the CEM are quite unique. It contains a large hydroelectricity base, around 70%. A third of this electricity depends on a large reservoir capacity, which is affected by severe weather variations which reduces water inflows into the system by up to 50%. As Colombia adopted the British regulatory scheme [29], some changes were introduced to account for structural differences between these systems. A complete description and evaluation of the CEM is presented by Larsen et al. [6] and Arango et al. [3]. Briefly, the results of a decade of operations seem satisfactory in terms of wholesale price, reliability of supply, technology evolution, and management development. However, price volatility and gains for the household sector are pending. Furthermore, black clouds are ahead with respect to the appropriate technology mix which

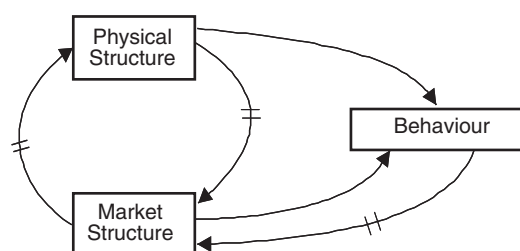


Fig. 1. Macro-structure dynamics of electricity systems.

influences price volatility and reliability of supply. Fig. 2 shows the evolution of the pool and long-term contract price, the aggregated stored water, the margin of the system (difference supply—demand over demand), and the South Oscillation Index (SOI).

An important aspect in the CEM is the influence of the macroclimate phenomenon called ENSO (El Niño South Oscillation). One of its indicators is the SOI and its occurrence significantly reduces the water inflow to the reservoirs, as well as many other sectors such as agriculture. The SOI index is used in the model to forecast the occurrence of ENSO. The opposite follows, “La Niña”, where the water inflow is higher than average. Consequently, with the ENSO and the fact that the system has approximately 70% of hydro generation capacity, in such phenomenon the prices increase considerably. In Fig. 2, the effect of the ENSO over the system during 1997–1998 is clear, the pool prices reached their maximum levels and the water reserves dropped considerably. Two coinciding factors have affected the CEM lately: a large fall in demand and a high water inflow after the ENSO 1997–1998. Fig. 2 shows that the capacity has increased and peak demand has fallen, creating an excess-capacity situation. This may drive prices down and discourage investments in new plants, with undesirable long-term consequences. These plants may not seem viable now but may be required for the next ENSO [6].

The principles behind the pool in the CEM are to provide long-term economic signals for both expansion of installed capacity and service reliability. In order to satisfy these principles CREG developed the capacity payment or capacity charge (CC) [31,32], and reservoir interventions when they reach certain limits.

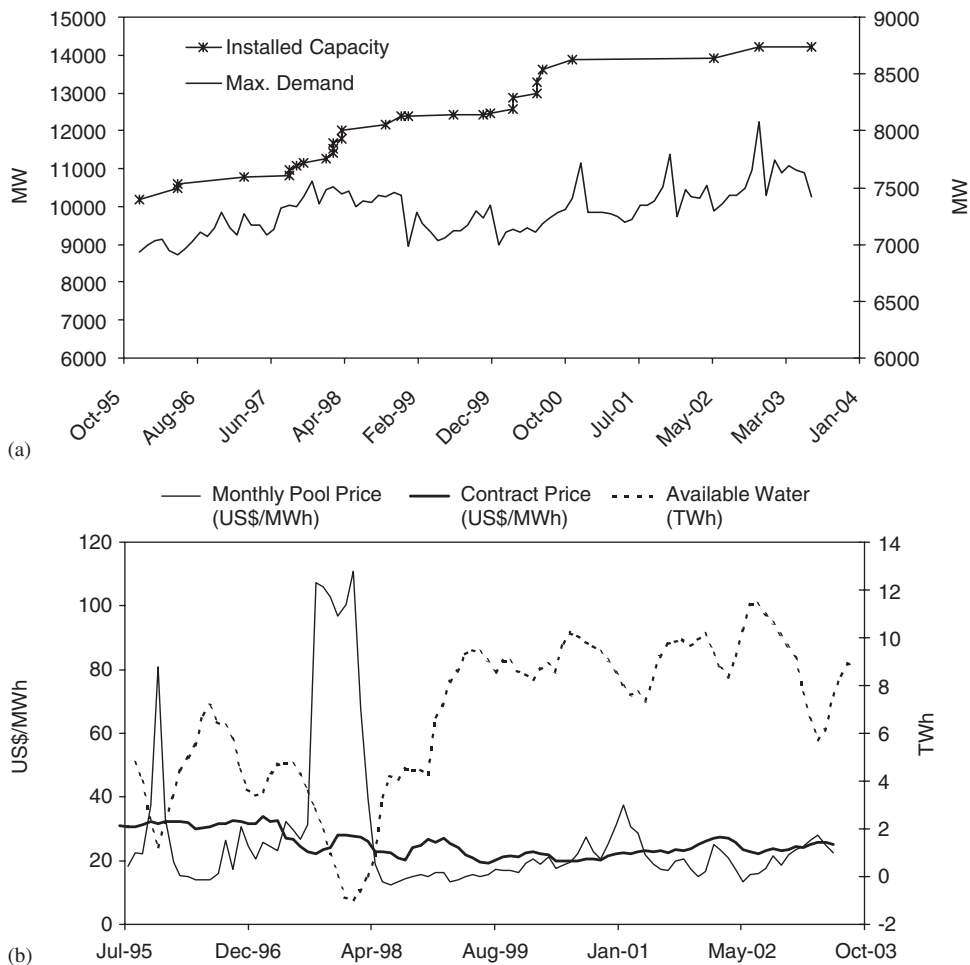


Fig. 2. (a) Installed capacity and maximum monthly power demand in Colombia. (b) The evolution of pool and contract electricity prices in Colombia from 1995 to 2002 (Source: [30]).

These limits are imposed on reservoirs to secure short-term reliability by intervening in the bid price of hydro-generators with water levels under the reservoir limits. The CC is a payment given to generators who provide, according to a predefined procedure,¹ reliability to the system. The reference value for the CC is estimated as the cost of installing 1 kW of an open cycle gas generator. The CC has been established at 5.25 US\$/kW. It is collected in the pool and becomes the lowest pool price. The CC is allocated by using a long-term model with specific conditions according to regulations: Resolutions CREG-077, CREG-082 and CREG-111 of 2000 [34].

In the region, Chile and Argentina have also included capacity payment mechanisms into their electricity systems, and Brazil has put them under consideration [35]. In Argentina, the payments are given to generators that contribute energy in the 90 h of the weekly peak demand period and is fixed at 10 US\$/MWh; In Chile, the payments are given to generators that contribute in the yearly peak demand period (May–September) and is also fixed at 5.25 US\$/kW/month (the same as Colombia).

The CC mechanism has been largely criticised in Colombia and has generated a number of complaints by different agents of the electricity market, especially for its effectiveness in providing real investment signals [11,13,14]. In fact, this paper explores alternative regulatory options to replace this mechanism. We present a summary of the criticisms based on [11,13]:

- i. The CC receivers do not have commitments as a counterpart of the payment. They receive the CC as extra income, without any guarantee of reliability. Thus, the demand does not guarantee supply in a crisis situation by the generators that receive the CC.
- ii. The CC allocations are based on simulation and optimisation modelling, with parameters that do not properly represent the system. The optimisation model was developed for a totally different environment, such as the system before deregulation with central dispatch.²

The time has arrived for reviewing the present regulation framework of the CEM. The transition may not be smooth and there might be some bumps along the way. Slightly bigger steps may now be required to incorporate schemes that have already been successful in similar industrial systems and also for adopting ideas experienced elsewhere. In this context, consultants and researchers have been called into examine the most troublesome issues affecting the system. A number of suggestions have been made. The rules relating to the security of supply are first on the agenda and several alternative solutions have resulted from discussions. Some of these are administrative and others are market-oriented solutions, the former are fairly interventionist and the latter more market oriented.

Alternative regulatory options need to be evaluated; simulations may help in this respect. It will be possible to assess the regulation's effect on capacity installation, system reliability, technology performance (plants with the same technology) and price trends. At the outset, five different alternatives were examined but we only report two in this paper. First, we present an administrative solution, where the regulator has an important role in guiding the market. Second, we present a market-oriented solution, where the regulator has a much lighter role and the system relies more on an option market.

2.2. Description of the system dynamic platform³

The model was built to address the problems facing the CEM. The main objective is to build a tool for policy analysis of alternative regulatory options. The previously described challenges led to the development of a system dynamics model, which has taken the form of a system dynamic platform [20]. The platform simulates the CEM from the regulator's point of view, and all the parameters were estimated from public data.

¹The CC is estimated based on an indicative plan developed and executed by the Ministry of Energy and Mines. A set of the worst scenarios is run to determine the "optimal" capacity payment and distributed among those generators that provide reliable energy [6,33].

²A recent study shows some tests of this optimisation model, by presenting some cases where the model does not converge to an optimal solution [36].

³The model was developed in a research project between the National University of Colombia, Medellín (www.unalmed.edu.co). COLCIENCIAS: "Instituto Colombiano para el Desarrollo de la Ciencia y la Tecnología Francisco José de Caldas", the Research Council in Colombia (www.colciencias.gov.co). ISA S.A.: The market operator and the largest grid company in Colombia (www.isa.com.co).

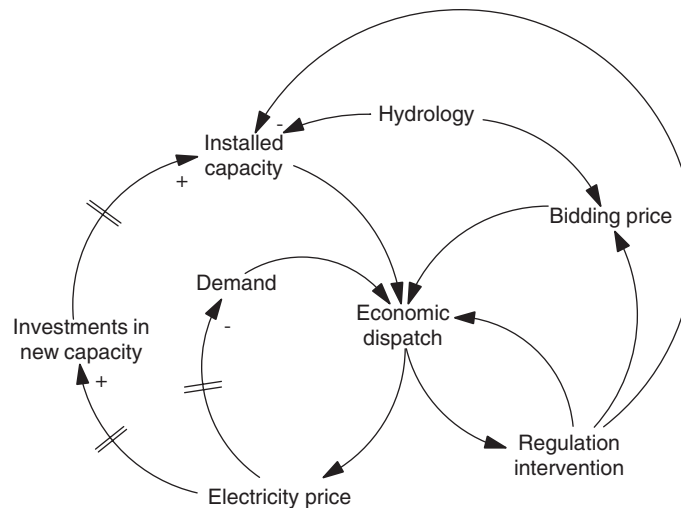


Fig. 3. General causal diagram of the system dynamics platform for policy analysis in the CEM.

It is designed at a relatively high level of aggregation compared to the other detailed production simulation models (see Appendix B in [37]). For example, the simulation is run with a monthly resolution, while the market operates on a daily basis.

The model is represented by the causal diagram shows in Fig. 3, which provides a general overview. Several feedback loops are taken into account. The investment loop takes into consideration the behaviour of investors and presents how investments take place in the model. The dispatch has the market price as an output and the price and the market development are taken by investors as a signal. Investment decisions will, after a delay, increase the installed capacity, which will change the topology of the system and therefore the economic dispatch.

The regulator follows the behaviour of the system by looking at economic dispatch development. Consequently, the regulator adjusts the long run market rules and investment incentives. The market rules determine the bidding price and the investment incentives influence the investors' behaviour and therefore the installed capacity. Since it determines the water availability, hydrology is an external factor, which influences both the bidding strategies and the installed capacity. Note that the price has an effect on demand. This effect is not taken into account as a consequence of the current market rules in the CEM as explained in the description of the model. Next, we present a more complete description of the different components of the model.

In the system, demand for electricity is set according to the scenarios defined by UPME [33].⁴ There are three defined scenarios: low, medium and high. The medium scenario was used as the baseline scenario of the simulations. The scenarios of demand define the growth for both energy and power. The model represents the electricity demand as the monthly energy required. The variations in demands over the hours in a day, over the days in a week are not taken into account, but the seasons of the year are considered.

The pool is a supply-side bidding market, which is a central electricity clearinghouse to dispatch generation to meet electricity demand. Generators participate in a post-price auction to get dispatched. Note that there is no demand-side bidding. The demand does not participate in the market, although it may be important as is presented in previous studies of electric utilities [26,38]. In case of shortage, contingent plans will be made and the market price will be set at the cost of the most expensive generating unit. Therefore, the model accounts shortages and the price will be increased according to the maximum bid.

The generation capacity is aggregated in the model into four technologies. These technologies are hydro, thermo, run of river and coal. Hydro represents all the hydro generators with reservoirs, which are modelled with their particular reservoir structure. Run of river aggregates the hydro capacity without reservoirs. Since

⁴UPME: Unidad de Planeación Minero Energético, Bureau of the Ministry of Mines and Energy in Colombia (www.upme.gov.co).

the marginal cost of the run of river technology is zero, because there is no water value without a reservoir, all the water inflow is dispatched for generation. Thermo and coal generation capacity use the average historical generation. The grid constrains the systems and creates imperfections in the market. In the model, we have simplified the grid constraints by imposing a minimum generation of each technology, independent of the economic dispatch.

The market price⁵ (MP) is determined by an economic dispatch that mimics the pool price mechanism of the Colombian electricity pool. The model estimates horizontally the summation of the supply curves for each technology to obtain the aggregated supply curve for the industry. Then, the demand equals the supply and the market is cleared. The supply curves were estimated as an average of all generators and all bids, using historic data since the beginning of the market. This data is aggregated by technology, season and by macroclimatic conditions. See the reference for details about supply curves [14]. Bids are expected to include not only the CC, but also the costs of generation. The strategic bidding behaviour of generators and the costs are taken into account within the supply curves.

An inverse process is followed in order to estimate the dispatch of every technology. Once the MP is calculated, the inverse side of the supply curve is taken for each technology. The resultant quantities of electricity are the generations of the technologies. Note the total generation of each technology will be the resultant quantity of the economic dispatch plus the minimum generation of the technology.

The investment behaviour in the model was represented with a real options approach. The real options approach captures the managerial flexibility of investment into the traditional net present value. Assuming a stochastic process for the price, the methodology allows to find the critical price (P^*). This means the price level at which the project under consideration can be justified for investment [39]. The P^* values were estimated externally by assuming that the price follows the geometrical Brownian motion and costs for the CEM [14]. The investors follow the state of the system to form expectations, by looking at pool price, investment incentives, macroclimatic conditions, and supply–demand balance. These are computed into the expected price (EP). The EP is therefore interpreted as the investors' expectations of market development.

The investment decision rule is a comparison between the P^* , the price at which the project is profitable, and the EP, the investors' expectation of market development. Thus, if $EP > P^*$, there is an investment on the project with lower P^* as long as it satisfies the construction delay. Therefore, the most profitable project will be the first to be built. We assume permits of all kinds are included with minimal time to enter. The model assumes investors are continuously monitoring the market development and price behaviour; therefore the decision rule is applied every month. The model satisfactorily replicates the investment decisions between 1996 and 2001 [14]. The available projects are the ones registered by the UPME, which vary from large-scale hydropower generation projects to multiple sizes of combined cycle co-generators (CCCTs).

The model simulates the water inflow to reservoirs and occurrences of the ENSO phenomenon. The water inflow is modelled using a stochastic autoregressive model depending on the regime, where the regime shows the occurrence of the ENSO phenomenon. The ENSO is modelled by backward sample techniques, which takes advantage of more than 100 years of data available for the phenomenon and replicates past occurrences. Both parts are extensively documented in Ref. [14].

The model satisfactorily replicates the historical market behaviour in terms of prices, generation, water level behaviour and investments as shown in Fig. 4. It also resembles the oscillatory pattern expected for electricity markets [16,37]; and IEA, 1999). Follow up researches have used this system dynamic platform. The Energy Institute at the National University of Colombia has used the model research projects such as the “Microworld for electricity investments in Colombia” [40], also adapted among others to Latin-American electricity markets [41], to explore the potential of wind technology in the CEM [42]. It has also been used in a number of Master Theses. Thus, the model has been tested in connection with other projects. The model was

⁵The MP is measured in US\$/MW (same as mills/kWh); it represents a monthly average price for electricity. The model was tested under a variety of conditions, where a reduction in the supply leads to a increase in the MP, and a gradual increase in the electricity demand leads to a gradual increase in the MP. Note also that the monthly average take into account the complexity of the price distribution over the hours in a month. Electricity prices in the Colombian have a wide range of variations, they tend to be higher in the peak hour (19.00h), than in the low load period. Thus, the model does not resemble the volatility due to the variations within a month.

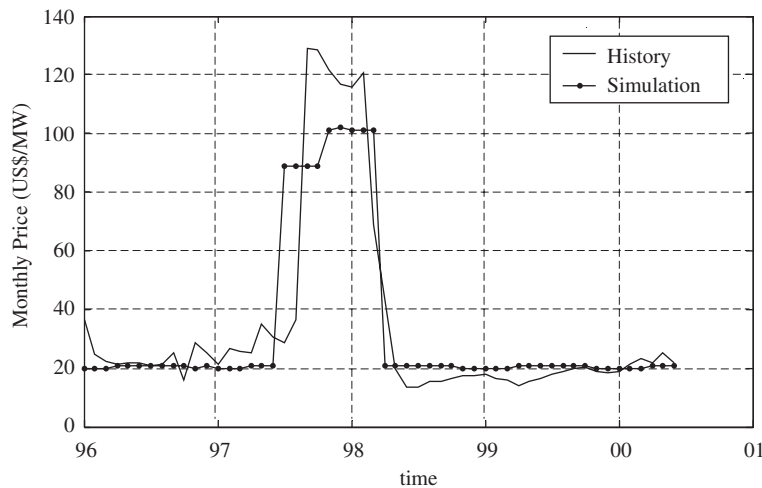


Fig. 4. Simulated vs. real pool price.

developed in Powersim,⁶ which includes connections to external data. To see the application of the model, we move on to show the simulation results of two alternative regulatory options.

3. Policy analysis and simulation results

To reiterate, innumerable alternative regulatory options could be applied to the CEM. Any of these could lead to a perfect market [43]. In this section we report the simulation and evaluation of two particular alternatives: availability payment and options market. The first presents an alternative methodology to pay the current CC. The second takes the alternative regulatory options suggested by the two external advisors of the CREG [11,13]. The model allows the use of different metrics for the evaluation of the regulatory options, which includes price variations (volatility), perceptual distribution of capacity for technology, and the cost of the system. Given that the bases for the regulatory options are different in structure, we present the metrics that we consider more appropriate for the evaluation.

3.1. Availability payments

Availability payments are presented as an alternative regulatory option to replace the CC. This alternative sets a methodology to allocate the resources collected in the pool (for reliability). The regulation consists of rewarding plants for being available when needed. The CC will be given to those plants which bid prices between the pool price and the price set by the threshold. The present mechanism to estimate the CC allocation would not be relevant under the proposed scheme. It is expected that some generators will not produce certain periods of the year, but they may be needed in the future. Thus, the generators that actually get the capacity payment will have to be available when needed, which provides short-term reliability. They will have to compete for this payment in the same way that they compete in the pool. In this sense, the availability payment acts as a pool for reliability. The regulation is expected to provide the same signal as the current CC does, since it is the same incentive, but it provides a market mechanism to allocate the CC payment.

Fig. 5 indicates how this regulatory option operates. With this regulatory alternative, the pool operates as usual in the CEM. The generators make daily bids for dispatch and they are ordered according to merit (cheaper plants are preferred over the more expensive ones). The power plants offer to the pool and the market build the supply curve shown in the figure. The cheapest plants will produce electricity up to the point that

⁶Powersim Constructor 5.1. See the webpage for details www.powersim.com.

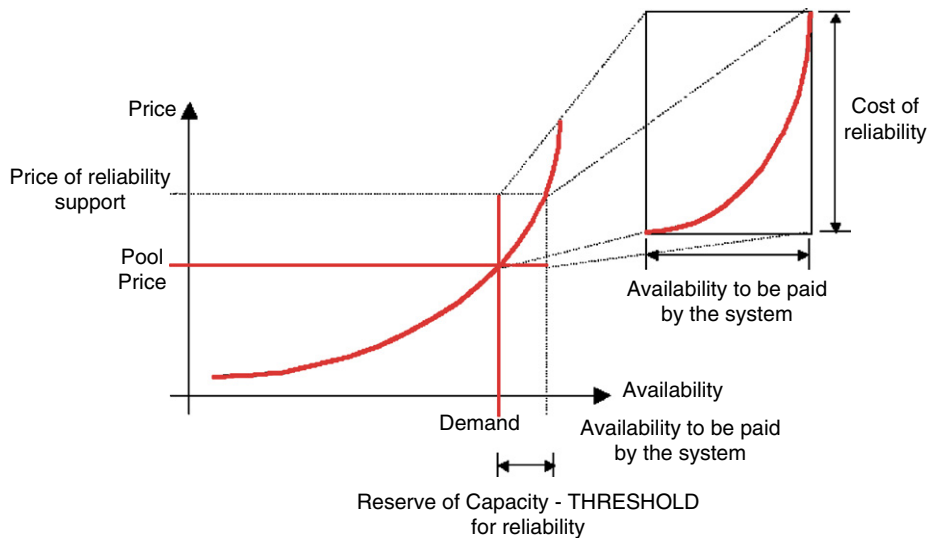


Fig. 5. Availability of payment up to an established threshold.

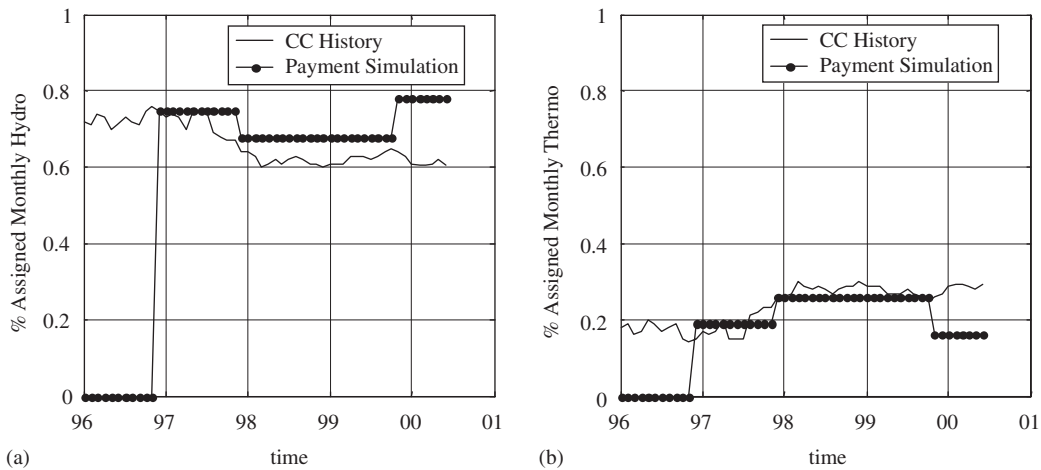


Fig. 6. Percentage of assignment for historic CC and availability payments to hydro and gas fired plants.

demand is satisfied. A threshold beyond the plants required for dispatch is established by the regulator, indicating which plants will also be rewarded. Thus, such payments represent the cost of reliability.

Fig. 6 shows both simulation results that would have happened if the scheme were applied during the previous years and also the historical market behaviour for hydropower technologies (see Fig. 6a) and gas fired plants (see Fig. 6b). In this case, a threshold of 30% over the electricity demanded, has been used. As can be seen, gas fired plants will benefit at the expense of hydroelectricity plants that possess large reservoirs during the ENSO period, and vice versa during the opposite phase of the ENSO.

Fig. 7 shows the percentage distribution reward for availability payment, according to technology where critical hydrology is present. As shown, the mechanism would benefit thermoelectricity plants over the hydroelectric, with more impact at the beginning of the applications of the alternative. However, the system behaviour tends to stabilise in the long run, with a portfolio with a higher percentage of hydro in the installed capacity.

The implementation of this alternative has a clear advantage. The only change that the system requires is the estimation of payments and distribution. Such estimations are similar to the way the pool works currently; therefore, the only new task for the regulator is to define the threshold of availability payment. Note that the

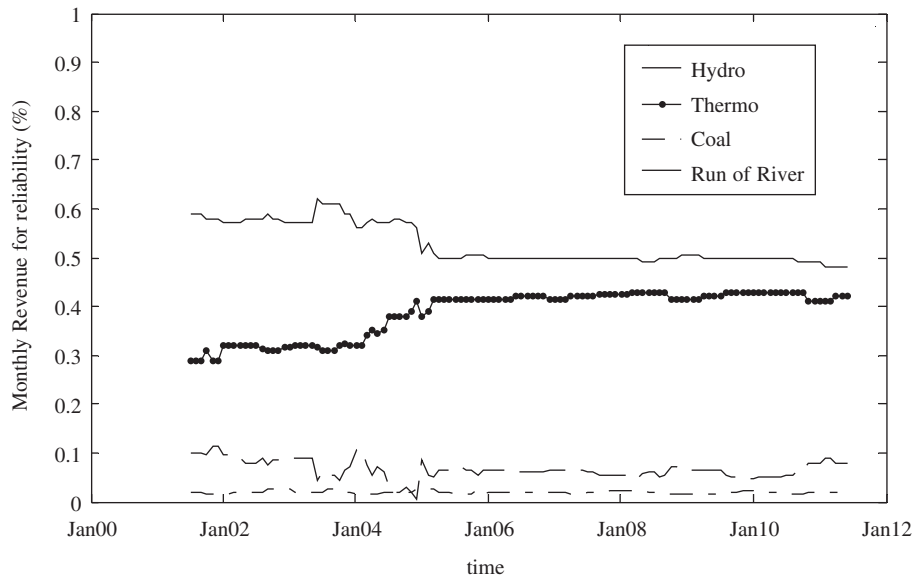


Fig. 7. Simulation of market split revenues according to technology.

generators which may be paid according to this alternative will provide firm energy to the system, since they must be available in the pool. The payments are the same as the CC and the mechanism to allocate the availability is very clear, since it is similar to the current pool. Thus, the alternative will provide transparency to the CEM, because it is a very clear form of defining the payments.

This alternative overcomes one of the CC problems, but there may still be questions about its real effectiveness on investment signals. A potential problem with this alternative is the short-term view; it may be possible for speculators to get extra profits with no guarantee for long-term investments. The alternative option is a proposed solution where the regulator still has a more active role in the market. The opposite regulatory option is market oriented and relies in the effectiveness of the markets.

3.2. An option market

A market-oriented solution to the problem of supply security is based on standard financial instruments designed for financial markets. The basic idea is to create financial and tradeable instruments intended to protect buyers from a sudden increase in price, at the expense of a premium which the buyer pays monthly. Inspired by CREG and COMILLAS [13,11], the proposal replaces the CC with a market mechanism that reflects real value by the interactions of the interest group in the reliability of the system. It includes both demand and supply, while the regulator only sets the market rules.

The financial instruments to be traded are *call options*, which give the buyer the option to buy energy in the pool at a given *strike price*. In compensation, the buyer must pay a *premium price* to the seller. The generators are the sellers and the demand represents the buyers. The generator must compensate the demand in case the pool price is higher than the strike price; the generators also have a commitment to provide the electricity agreed upon in the option. The options are traded in a public auction, where all the demand must be covered. The market is expected to define the price that electricity customers are willing to pay for reliability by clearing demand and supply. Thus, the price should move up and down according to the balance of demand and supply. Note that there is no need for external models that may complicate the market rules. In this case, each generator has to discover how to construct a portfolio of plants to guarantee power supply to its customers. In this manner the market will find the appropriate technology mix to reduce the probability of electricity blackouts. The regulatory option also considers the elimination of minimum operating levels, since the market will ensure the reliability of the market. The electricity price received by

generators is

$$EP = PR + [\min(PP, SP) - CC],$$

where PR is the option price, PP is the electricity pool price, SP is the strike price and CC is the current capacity charge; all measured in US\$/MW. Note that there is no payment for the CC, instead the generators receive the extra revenue due to the option price or premium (PR). As in the model we use the historical supply curves to estimate the pool price which included the CC, the CC must be subtracted. The strike price and premium would be set by the new market.

Given that there is no reference market, we assumed them to be constant, so built a base case scenario and performed a sensitivity analysis. The base case scenario was constructed in discussions with people from the electricity industry in Colombia, so we have obtained values which are considered reasonable. The base case scenario considered a strike price of 55 US\$/MW and a premium of 15 US\$/MW and simulations were run under this and other scenarios. We now present the simulation results of the base case scenario. Fig. 8 shows the volatility of both historical monthly average electricity prices and simulation results during the 1st years of the deregulation, assuming that the options market was in operation instead of the CC. An options market manages to spread rewards to producers, reducing peaking prices during periods of low water inflow, like during El Niño periods. Customers benefit for the same reason, and therefore, the pool price would experience a reduction in volatility.

In general, we observed the options market would have benefited customers in the long run by smoothing the effect of El Niño. Fig. 9 shows revenues of generators according to technology, for both historical behaviour and as if the options market had been in operation. The historic tendency does not change, but also some compensation may have occurred between hydro- and thermo-electricity plants.

Finally, Fig. 10 shows that the expected future price evolution, under critical conditions, operates basically as was illustrated with historical data. Volatility is largely reduced, company revenues are more evenly distributed, and average prices are slightly lower. Note that this option redistribute the total cost of the system, by spreading revenues across technologies, by reducing revenue for the hydro plants during the concurrence of El Niño and increasing it for the thermo plants. This situation is more reasonable, since during such phenomenon the hydro plants do not provide reliability to the system.

One of the advantages of this alternative is that active demand will determine the price for reliability. The reliability of the system will be the result of a market mechanism that may be difficult to implement. The application of this alternative has strong implications for the actors in the market since they have to update

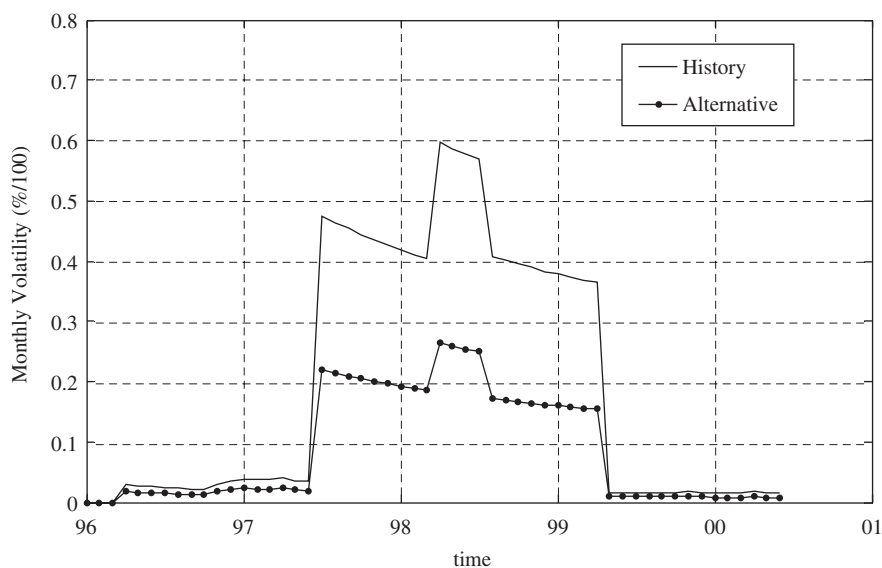


Fig. 8. Volatility during the period 1996–2000, historical vs. options market simulation.

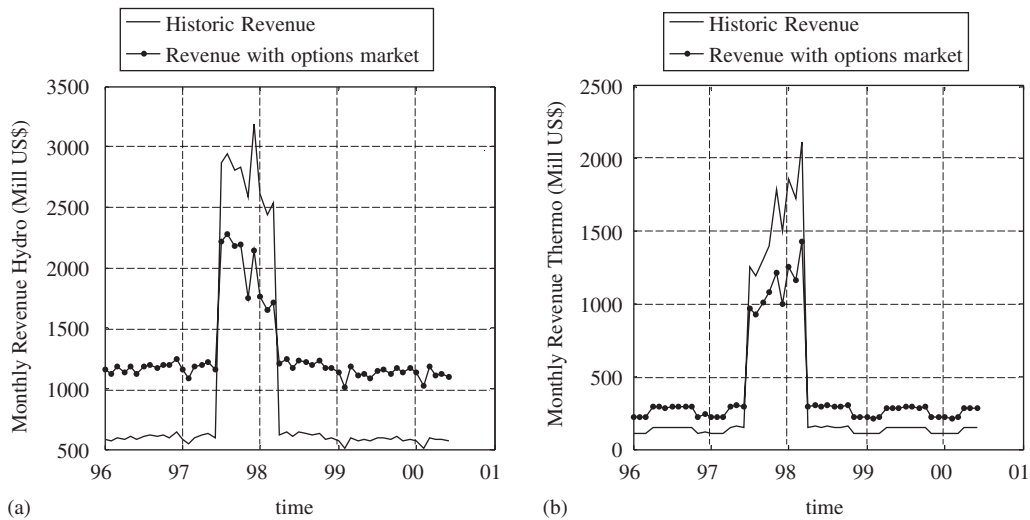


Fig. 9. Revenue for plants, according to technology ((a) is hydro based; (b) is gas based). Historical revenues are compared against options market.

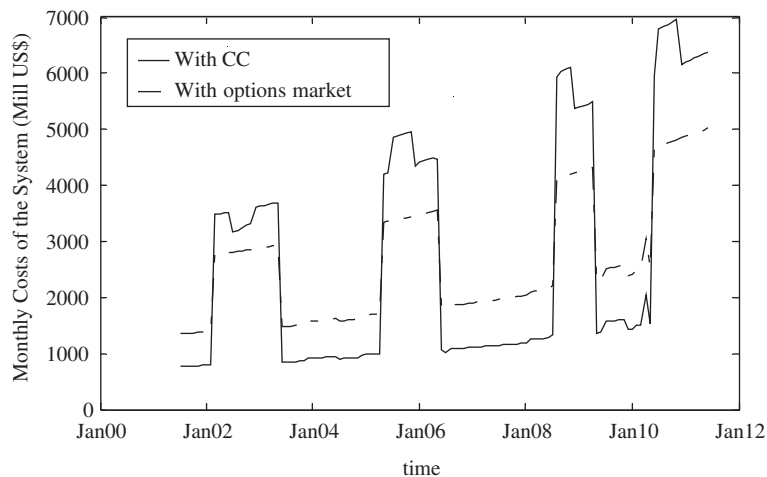


Fig. 10. Simulation of the future price evolution under a critical hydrology scenario.

their knowledge of such financial tools and the regulator has to observe carefully any exercise of market power. There is a need to define many details and very important elements, in particular the design of the pool, the degree to which the market would be voluntary or mandatory and the frequency of the pool. Previous reports [11,13], as discussed before, have developed different proposals to overcome these problems.

Other regulatory alternatives have been evaluated and others are being examined. Given the focus of the paper, which is basically methodological, the author believes that the above examples illustrate the approach, without going into further detail. One alternative regulatory option takes the CC element and changes the way of payment, the second presents a liberalised option that follows the suggestion of external advisors. The next section provides a discussion and concludes the paper.

4. Final discussion

The CC will expire in 2006 and different alternative regulatory options have already been under discussion. These changes are necessary as the regulation should evolve and adjust to the new requirements. Colombia

cannot fall in to the same electricity crisis as Chile in 1998, Brazil in 2001 and recently Argentina. The electricity system market must update the deregulation. We have examined alternative regulation schemes for the CEMs by using simulation. The method applied for evaluating each alternative over time consists of trying to establish the evolution of the physical and topological structure. In this way, each alternative is assessed with respect to its reliability and price to customers, including volatility.

We have presented two alternative regulatory options. We veer towards the creation of the option markets. As option market will make the system state-of-the-art. One disadvantage of an option market may be the limited international experience. Methodologies such as experimental economics and simulations could bring more detailed and critical evaluation of a particular option. Moreover, during the implementation phase the regulatory framework will need continuous improvement. Simulation results show the system's behaviour may be improved by introducing modifications to the present schemes in operation, eliminating the CC and putting one of the alternatives, being examined, in place. In almost all cases price system uncertainties are drastically reduced, which benefits both customers and investors. In some cases volatilities are reduced by up to 40%. Furthermore, even under the most critical scenarios few blackouts were observed. Further research is required for the selection process.

The policies were evaluated with a system dynamic simulation model. The system dynamics approach takes the form of platforms for policy evaluation and provides a decision support tool which may be used by the market regulator. The regulator may use the developed model to assess alternative market schemes, such as the ones that we have evaluated. Note the model needs adjustment under particular alternative regulatory schemes, i.e. if the alternative policy implies a change in the market structure, this change must be included in the modelling platform.

The analysis of the alternatives may look to be in favour of thermo-electricity, which is not the exact intention here. We have observed that the signals should be moved for two criteria: reliability of the system and sustainability. In this work, we focused on reliability. There is a need for further research on the effect of the mechanisms over alternative energies, such as wind generators and small-scale hydro electricity plants.

The simulation model is not intended to be a finished product. Improvements could be made by including the grid in more detail, by taking a different approach to the load curve for demand during the day, or by using a different investment behaviour model. Follow up research has used the system dynamic platform presented in this paper. The Energy Institute at the National University of Colombia has used the model in different research projects, such as the Microworld for electricity investments in Colombia [40] and it was also adapted by some Latin-American countries [41], to explore the potential of wind technology in the CEM [42]. It has also been used in a number of Master Theses. Thus, the model has been tested in connection with other projects.

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